

Name: _____

Symbolic Logic

Practice Second Examination

Real exam time limit: (50 minutes)

Answer all questions

Scores

Part A, Satisfying Formulas: _____/30

Part B, Interpretations: _____/30

Part C, Derivations: _____/40

Total: _____/100

Part A. Satisfying Formulas (30 points) Use the following domain and interpretation to determine whether the provided expression is true or false.

D: Humans in the following diagram

B(1): is bored.	l: larrie.
C(1): is a cab driver.	r: ralph.
D(1): is a dog trainer.	
H(1): is hirsute.	
W(1): is wary.	

1. True or False: “Everything is human.”
2. True or False: “All bored cab drivers are wary.”
3. True or False: If Larrie and Ralph are human, then both are hirsute.
4. True or False: $\sim \forall x((Dx \ \& \ Wx) \vee Bx)$
5. True or False: $\forall x(Wx \rightarrow \exists x(Bx))$

Part B. Provide an Interpretation (30 points)

6. Provide an interpretation in a small finite domain that satisfies the following set of sentences (i.e. makes them all true). **Be sure to provide a complete interpretation, include the domain and explain your predicates and constants.**

$\exists x(Ax \vee Bx), \forall x(\sim Bx \vee Ax), \sim \exists x(Ax \ \& \ Bx)$

7. Provide an interpretation in a small finite domain that satisfies the following set of sentences (i.e. makes them all true). **Be sure to provide a complete interpretation, include the domain and explain your predicates and constants.**

$$\exists x Fx \rightarrow \exists y Gy, \sim \forall x Fx \vee \exists y Gy$$

Part C. Derivations (40 points)

Provide derivations that show that the following arguments are valid. (You will receive points for starting a derivation in a promising direction, even if you do not finish it. You will receive more points for an incomplete derivation that is correct as far as it goes than for a “complete” derivation that misuses the rules.)

8. $\sim \forall x Fx \rightarrow \sim \exists y Gy$; $\forall x Fx \vee \exists y Gy$; SHOW $\forall x Fx$.

8

Space for Working, Rewriting Answers, Etc.

